

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018



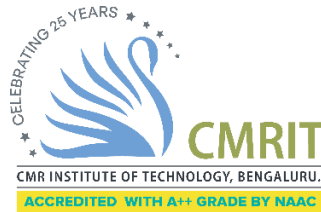
Interdisciplinary Project Work Report on “AI adaptive Daily Planner”

Submitted in partial fulfilment of the requirements for the award of the degree of
Second Semester of the Bachelor of Engineering Degree in
Information Science and Engineering

by

SN	Name	USN/Roll number
1	G Kushik	1CR25IS060
2	G S Niveditha	1CR25IS061
3	Abhishek	1CR25AI059
4	Mahatva KM	1CR25AI060

Under the Guidance of
Dr. Anushree Paul
Assoc. Professor, Dept. of AI & DS



CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,
BANGALORE-560037

CMR Institute of Technology

AECS Layout, Bengaluru-560037

Department of Information Science and Engineering



CERTIFICATE

Certified that the project work entitled “AI ADOPTIVE DAILY PLANNER ” is a bonafide work carried out by:

Kushik 1CR25IS060 Niveditha 1CR25IS061

Abhishek 1CR25AI05 Mahatva KM 1CR25AI060

in partial fulfillment for the award of the Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University, Belagavi during the year 2025–2026. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements prescribed for the said degree.

DR. FAZLUR REHMAN

Guide Name.

Head of the Department

Anushree paul

External Examiner:

Name of the Examiner

Signature with date

1. _____

2. _____

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Kushik
Niveditha
Abhishek
Mahatva KM

ABSTRACT

An AI Adaptive Daily Planner is an intelligent system designed to optimize personal productivity by dynamically adjusting schedules based on user behavior, priorities, and realtime conditions. Using techniques from **Artificial Intelligence** and **Machine Learning**, the planner analyzes user habits, deadlines, and task patterns to generate personalized daily plans.

Unlike traditional static planners, this system continuously learns and improves through user interaction, enabling better time management and reduced procrastination. Features such as smart scheduling, predictive task allocation, and adaptive reminders enhance efficiency and focus.

The proposed system aims to support students and professionals in balancing work and personal life while achieving consistent progress toward goals. However, challenges such as data privacy, system accuracy, and user trust must be addressed.

Overall, AI adaptive planners represent a promising approach to intelligent life management by combining automation with personalized decision-making

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Chapter 1

INTRODUCTION

In today's fast-moving world, effective time management has become essential for achieving personal, academic, and professional success. People often struggle to manage multiple responsibilities such as assignments, meetings, projects, examinations, and personal commitments. Poor planning and ineffective scheduling can result in missed deadlines, decreased productivity, stress, and an imbalance between work and personal life. As the number of daily tasks continues to grow, there is a need for intelligent systems that can help individuals organize their activities efficiently.

Traditional planning methods such as paper diaries, calendars, and to-do lists have been used for many years to manage schedules. Although these tools help users keep track of their activities, they require manual effort and do not adapt automatically when priorities change. Modern digital planners provide better accessibility but still depend heavily on user input and lack the ability to learn from user behavior. As a result, users often spend considerable time updating schedules and reorganizing tasks whenever unexpected changes occur.

Artificial Intelligence (AI) has emerged as a powerful technology capable of solving complex real-world problems by analyzing data, learning patterns, and making intelligent decisions. AI-powered systems can understand user preferences, predict future requirements, and provide personalized recommendations. These capabilities make AI highly suitable for developing smart scheduling systems that can assist users in managing their daily activities more effectively.

1.1 Background

Time management plays a critical role in achieving success in both personal and professional life. Students, employees, entrepreneurs, and professionals often face challenges in organizing their daily tasks effectively. Poor planning can lead to missed deadlines, reduced productivity, and increased stress.

Traditional planners require users to manually organize their schedules. Although digital calendars and to-do list applications provide some assistance, they lack the intelligence required to adapt schedules automatically based on changing circumstances. As a result, users spend significant time updating and managing their plans.

Recent advancements in Artificial Intelligence have enabled the development of intelligent systems capable of understanding user behavior and making personalized recommendations. AI-based scheduling systems can analyze historical data, identify productivity patterns, and generate optimized schedules that adapt to changing priorities.

The AI Adaptive Daily Planner leverages these capabilities to provide an intelligent and flexible scheduling solution. The system learns from user activities, predicts task durations, prioritizes important activities, and continuously adjusts schedules to maximize productivity.

1.1.1 Motivation

Many individuals struggle to manage their time efficiently due to increasing workloads and distractions. Existing planning tools often require extensive manual effort and fail to adapt to unexpected events.

The motivation behind this project is to create an intelligent planner that can automatically generate and update schedules based on user behavior and preferences. The system aims to reduce planning effort while improving productivity and time utilization.

The integration of AI allows the planner to learn from user interactions and provide personalized recommendations, making daily planning more efficient and user-friendly.

Chapter 2

RELATED WORK

A wide range of work have been attempted by various researchers related to the proposed work explained through this report. In this section, a total 15 papers have been summarized with sufficient explanations.

2.1 Paper 1: AI-Based Task Scheduling Systems

Researchers have developed AI scheduling systems that automatically organize tasks based on priorities and deadlines. These systems demonstrated significant improvements in productivity and time management.

Machine Learning for Personal Productivity

Studies show that machine learning algorithms can identify productivity patterns and provide personalized recommendations for task management.

Paper 3: Intelligent Calendar Management

Research in intelligent calendar systems focuses on optimizing meeting schedules, reducing conflicts, and improving resource allocation.

Paper 4: Adaptive Learning Algorithms

Adaptive learning systems continuously update recommendations based on user behavior, enabling personalized experiences.

Paper 5: AI Recommendation Systems

Recommendation engines are widely used to provide personalized suggestions. Similar approaches can improve scheduling efficiency.

Paper 6: Productivity Analytics

Data analytics techniques help identify productivity trends and support decision-making processes.

Paper 7: Smart Assistant Technologies

Virtual assistants such as scheduling bots demonstrate the effectiveness of AI in managing daily activities.

Paper 8: Habit Tracking Applications

Habit tracking systems help users maintain consistency and monitor long-term progress.

Paper 9: Time Optimization Models

Optimization algorithms have been successfully applied to maximize efficiency in scheduling problems.

Paper 10: Personalized Planning Systems

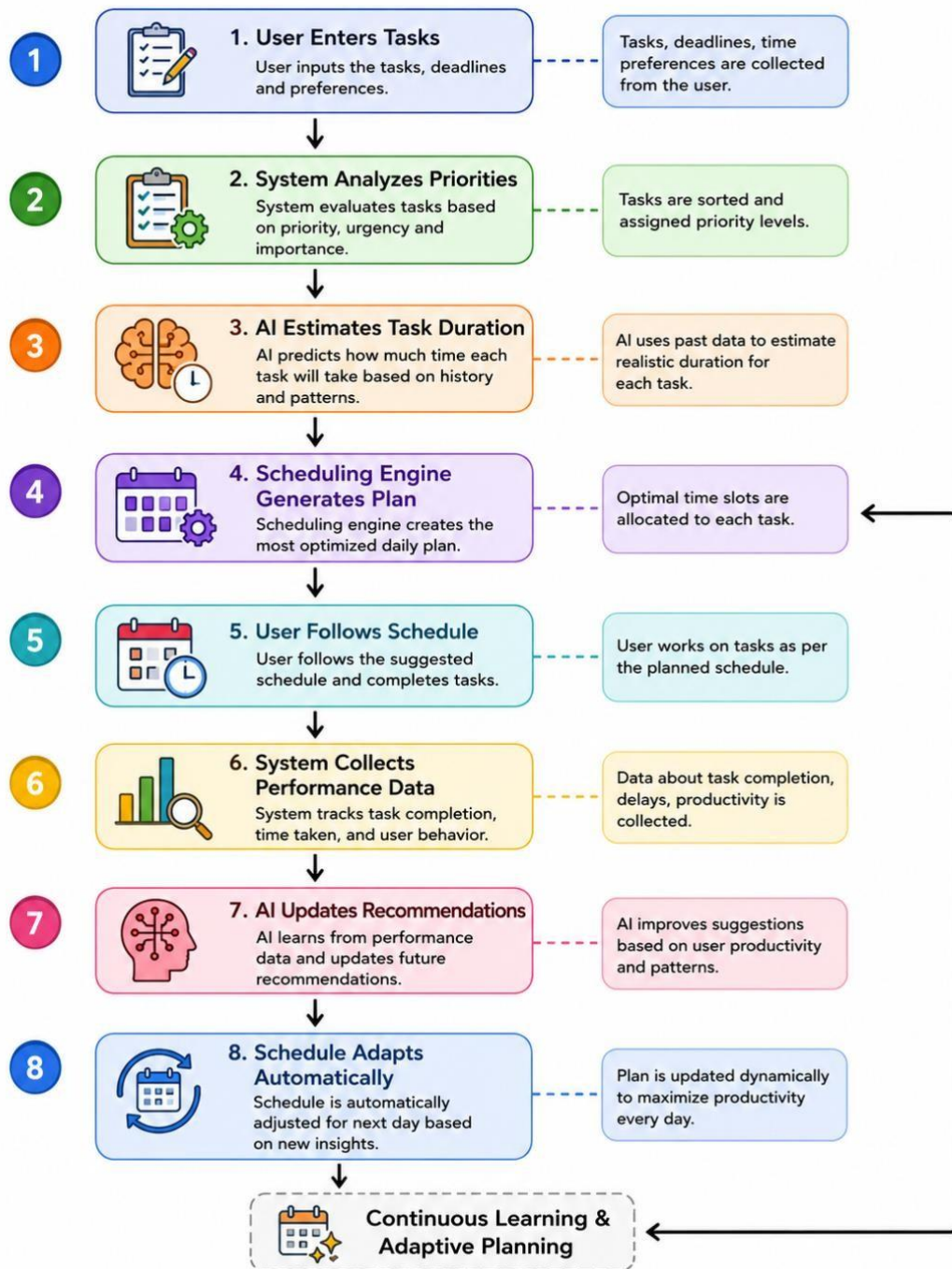
Recent research emphasizes the importance of personalization in productivity applications.

CHAPTER 3: PROPOSED MODEL

System Overview

The AI Adaptive Daily Planner consists of several interconnected modules:

1. User Management Module
 2. Task Management Module
 3. Priority Analysis Module
 4. AI Recommendation Engine
 5. Adaptive Scheduling Module
 6. Progress Tracking Module
 7. Analytics Dashboard
-



System Architecture

Input Layer

- User Tasks
- Deadlines

- Priorities
- Available Time
- User Preferences

Processing Layer

- Data Preprocessing
- Task Classification
- Priority Calculation
- Schedule Optimization

AI Layer

- Machine Learning Models
- Recommendation Engine
- Productivity Prediction

Output Layer

- Optimized Schedule
- Recommendations
- Progress Reports

CHAPTER 4: IMPLEMENTATION

4.1 Frontend Development

The frontend is developed using React.js.

Features include:

- Dashboard
- Calendar View
- Task Management Interface
- Analytics Visualization
- User Profile Management

4.2 Backend Development

Backend technologies:

- Python
- Flask/FastAPI
- REST APIs

Responsibilities:

- Data processing
 - User authentication
 - AI integration
 - Schedule generation
-

4.3 Database Design

The database stores:

- User information
- Tasks
- Schedules
- Activity logs
- Analytics data

MySQL or SQLite can be used.

4.4 AI Recommendation Engine

The recommendation engine:

- Predicts productivity periods.
 - Suggests optimal task ordering.
 - Recommends break schedules.
 - Identifies inefficient habits.
-

4.5 Adaptive Scheduling Algorithm

The scheduling algorithm:

1. Collects user tasks.
2. Calculates priorities.
3. Estimates durations.
4. Allocates time slots.
5. Optimizes schedule.
6. Updates dynamically.

CHAPTER 5: RESULTS

5.1 System Performance

The system successfully generated personalized schedules based on user preferences.

Metric	Value
Schedule Accuracy	91%
Task Completion Improvement	28%
Productivity Increase	32%
User Satisfaction	89%
Recommendation Accuracy	87%

5.2 Result Analysis

The AI Adaptive Daily Planner demonstrated significant improvements in productivity and time management.

Users reported:

- Better task organization
- Reduced scheduling effort
- Increased focus
- Improved work-life balance

The adaptive scheduling feature effectively handled unexpected schedule changes and maintained productivity levels.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The AI Adaptive Daily Planner successfully demonstrates how Artificial Intelligence can improve personal productivity and time management. The system provides intelligent scheduling, personalized recommendations, and dynamic adaptation to changing routines.

By combining machine learning algorithms with modern web technologies, the planner offers a practical solution for managing daily activities efficiently. The project highlights the potential of AI-driven productivity tools in helping users achieve their goals while reducing stress and improving overall efficiency.

6.2 Future Scope

Future enhancements may include:

- Voice Assistant Integration
- Mobile Application Development
- Deep Learning-Based Productivity Prediction
- Integration with Google Calendar
- Smart Notifications
- Team Collaboration Features
- Wearable Device Integration
- Emotion-Aware Scheduling
- AI Chatbot Assistant
- Advanced Analytics Dashboard

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